

C++20 MESH REDUCTION · GAME-ENGINE IMPORT PIPELINES

Fewer triangles. Measured transitions.

An independent visual audit turns mesh reduction into a measurable search. This board follows real meshes through all eight levels of their recorded LOD chains.

Geometry, with an error budget.

Propose a simpler mesh. Check it against the previous level and the original. Keep the smallest valid chain found.

FROZEN MESH CORPUS 120 30 rocks · 30 organics 40 manufactured · 20 complex scans	DIRECT COVERAGE / PILOT 87.68 SCORE +31.98 points over native QEM 8 development assets · unchanged gates	FULL CORPUS / SMOKE 90.78 SCORE 120 / 120 completed · zero failures Progressive coverage · smaller work budget	INTEGRATION PATHS 3 chain modes Direct · progressive · hybrid Rebuild vertices or reuse source buffers
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01 / ACTUAL OUTPUT GEOMETRY

From source to 16 pixels.

Four examples from the same direct coverage run. Every tile is an exported mesh, with its recorded triangle count.

DISPLAY **Clay** Wireframe Silhouette SCALE **Enlarged** Target px [Reset views](#)

Enlarged for inspection. Same camera and source scale across each row. Drag a mesh to rotate its chain; click a tile to inspect it.

01 / ROCK

Moon rock

A compact surface becomes a coarse silhouette.

L0D 032.0 px

L0D 129.0 px

L0D 226.3 px

L0D 323.8 px

L0D 421.5 px

L0D 519.5 px

L0D 617.7 px

L0D 716.0 px



3,304 tris
UNCHANGED SOURCE



66 tris
✓ Δ 1.83 / 2.00



60 tris
✓ Δ 1.41 / 2.17



54 tris
✓ Δ 1.27 / 2.33



48 tris
✓ Δ 2.29 / 2.50



44 tris
✓ Δ 1.41 / 2.67



40 tris
✓ Δ 1.50 / 2.83



36 tris
✓ Δ 1.21 / 3.00

−98.9% at LOD 7

3,304 → 36 tris

02 / ORGANIC

Quiver tree trunk

Fine surface detail is outside this coverage-only contract.

L0D 032.0 px

L0D 129.0 px

L0D 226.3 px

L0D 323.8 px

L0D 421.5 px

L0D 519.5 px

L0D 617.7 px

L0D 716.0 px



17,978 tris
UNCHANGED SOURCE



360 tris
✓ Δ 1.06 / 2.00



324 tris
✓ Δ 0.96 / 2.17



292 tris
✓ Δ 0.96 / 2.33



108 tris
✓ Δ 2.47 / 2.50



108 tris
✓ Δ 0.00 / 2.67



108 tris
✓ Δ 0.00 / 2.83



108 tris
✓ Δ 0.00 / 3.00

−99.4% at LOD 7

17,978 → 108 tris

03 / MANUFACTURED

Metal stool

Thin supports reveal the cost of aggressive distance LODs.

L0D 032.0 px

L0D 129.0 px

L0D 226.3 px

L0D 323.8 px

L0D 421.5 px

L0D 519.5 px

L0D 617.7 px

L0D 716.0 px



6,532 tris
UNCHANGED SOURCE



225 tris
✓ Δ 1.71 / 2.00



202 tris
✓ Δ 1.46 / 2.17



182 tris
✓ Δ 1.27 / 2.33



126 tris
✓ Δ 2.47 / 2.50



126 tris
✓ Δ 0.00 / 2.67



126 tris
✓ Δ 0.00 / 2.83



126 tris
✓ Δ 0.00 / 3.00

−98.1% at LOD 7

6,532 → 126 tris

04 / COMPLEX SCAN

Bell X-1

Shared positions reduce together while attribute wedges stay separate.

L0D 032.0 px

L0D 129.0 px

L0D 226.3 px

L0D 323.8 px

L0D 421.5 px

L0D 519.5 px

L0D 617.7 px

L0D 716.0 px



99,586 tris
UNCHANGED SOURCE



1,992 tris
✓ Δ 1.06 / 2.00



1,792 tris
✓ Δ 1.06 / 2.17



1,612 tris
✓ Δ 1.50 / 2.33



1,462 tris
✓ Δ 2.47 / 2.50



1,462 tris
✓ Δ 0.00 / 2.67



1,462 tris
✓ Δ 0.00 / 2.83



1,462 tris
✓ Δ 0.00 / 3.00

−98.5% at LOD 7

99,586 → 1,462 tris

Reading the tiles: Δ is the audited upper error against the previous LOD, followed by its pixel limit. Each reduced mesh also passed its direct-to-source limit. LOD 0 is unchanged.

Neutral clay uses geometric face normals, with no textures. Presentation angles are not the audit cameras. Enlargement exposes detail that was never required at 16–32 pixels.

02 / COMPARABLE COVERAGE SCORES

The gain came from shared positions.

Simplifying duplicated positions together unlocked large reductions in meshes with separate attribute wedges. UVs and other vertex attributes remain separate.

Native QEM		55.69
meshoptimizer		78.37
Fast Quadric		82.73
CGAL probabilistic *		84.28
Coupled positions		87.68

Same 8-asset development pilot, direct chain policy and visual gates. External adapters propose geometry; our evaluator accepts or rejects it. CGAL rejected one nonmanifold asset, which remains in its score denominator.

SCORE = 100 × (1 − category-balanced mean retained triangle ratio). LOD 0 is excluded. A score of zero means no reduction. Scores from different profiles or scenarios are not a ranking.

03 / WHAT THE EXPERIMENTS SAY

Progress, and the next hard problem.

120/120

CORPUS SMOKE

The pipeline runs across the corpus.
All assets completed in 16.9 minutes with no failures or unchanged-chain fallbacks. This uses a progressive policy and the small camera set.

1.08×

SIMD KERNEL

Faster in isolation; little change end to end.
The distance-transform fixture improved about 8%. Full pilot timing changed about 0.5%, which does not establish a useful speed gain. Output hashes stayed identical.

5.76

NORMALS SCORE

Attribute-aware reduction is the priority.
Strict normal preservation allows little reduction: three of eight chains stayed unchanged. Strong coverage results do not establish strong shading fidelity.

THE VISUAL CONTRACT 32 → 16 px · 8 levels Examples use a 2 → 3 px transition curve. Source-error budgets are checked separately. The same bounds and scale are used throughout each chain.	THE AUDIT BEHIND THESE EXAMPLES 12 orthographic + 4 perspective views Coverage is checked at 4× sampling with refinement to 8×. Search uses a separate camera set. These are finite sampled views, not a universal error guarantee.	WHAT REMAINS OUTSIDE THIS BOARD Full-quality corpus audit The default 642 + 64 camera preset has not been run across the full corpus. Texture images, opacity, skinning and morph targets are outside this version.
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